

Optimization Midterm 2025 Autumn

December 17, 2025

1 Optimality Conditions

1. A useful property for symmetric definite positive matrices: Let $A \in S_d^{++}$ then we have the following inequality for all $z \in \mathbb{R}^d$:

$$\lambda_{\min}(A)\|z\|^2 \leq \langle Az, z \rangle \leq \lambda_{\max}(A)\|z\|^2$$

Remark: This is very useful when dealing with quadratic forms. Also note that this is very useful to prove the second order conditions for optimality. (i.e. if the Hessian $\nabla^2 f(x)$ is positive definite at a critical point x^* , then x^* is a local minimum.)

2. **First Order Optimality Condition:** Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a differentiable function. A necessary condition for $x^* \in \mathbb{R}^d$ to be a local minimum of f is that the gradient at that point is zero:

$$\nabla f(x^*) = 0$$

3. **Second Order Optimality Condition:** Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a twice differentiable function. Let x^* be a critical point of f (i.e. $\nabla f(x^*) = 0$). Then:

- If the Hessian $\nabla^2 f(x^*)$ is positive definite, then x^* is a local minimum of f .
- If the Hessian $\nabla^2 f(x^*)$ has 2 eigenvalues of opposite signs, then x^* is a saddle point of f .
- If the Hessian $\nabla^2 f(x^*) \in S_d^+$ but is not positive definite, then x^* could be a local minimum or a saddle point (inconclusive).

Remark

Some useful multicalculus rules:

- **Product rule:** If $u, v : \mathbb{R}^d \rightarrow \mathbb{R}^d$ are differentiable functions, then:

$$D_x \langle u, v \rangle = \langle D_x u, v \rangle + \langle u, D_x v \rangle$$

- **Chain rule:** If $f : \mathbb{R}^m \rightarrow \mathbb{R}$ is a differentiable function and $g : \mathbb{R}^d \rightarrow \mathbb{R}^m$ is a differentiable function, then:

$$\underbrace{\frac{\partial f}{\partial x}}_{n \times 1} = \underbrace{\frac{\partial g}{\partial x}}_{n \times m} \cdot \underbrace{\frac{\partial f}{\partial g}}_{m \times 1}$$

2 Convexity

- 1. Tangent plane property of convex functions:** Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a differentiable convex (α strongly convex) function. Then for all $x, y \in \mathbb{R}^d$, we have:

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$$

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\alpha}{2} \|y - x\|^2$$

This means that the function lies above its tangent plane (plus a quadratic term for strong convexity) at any point.

- 2. Monotonicity of the gradient** Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a differentiable convex (α strongly convex) function. Then for all $x, y \in \mathbb{R}^d$, we have:

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq 0$$

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq \alpha \|y - x\|^2$$

This means that the gradient of a convex function is a monotone operator (stronger monotonicity than a quadratic function for strongly convex functions).

- 3. Characterization of convexity via Hessian:** Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a twice differentiable function. Then:

- f is convex if and only if its Hessian is positive semi-definite for all $x \in \mathbb{R}^d$:

$$\nabla^2 f(x) \in S_d^+, \quad \forall x \in \mathbb{R}^d$$

- f is α -strongly convex if and only if its Hessian's smallest eigenvalue is bounded below by α for all $x \in \mathbb{R}^d$:

$$\lambda_d(\nabla^2 f(x)) \geq \alpha, \quad \forall x \in \mathbb{R}^d$$

- If the Hessian is positive definite for all $x \in \mathbb{R}^d$, then f is strictly convex, but the converse is not true. (e.g. $f(x) = x^4$ is strictly convex but its Hessian is zero at $x = 0$.)

- 4. Segmentation function:** Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a differentiable function. Define the segmentation function $\phi : [0, 1] \rightarrow \mathbb{R}$ as:

$$\phi(t) = f(x + t(y - x)), \quad t \in [0, 1]$$

$$\phi'(t) = (y - x)^T \cdot \nabla f(x + t(y - x)) = \langle \nabla f(x + t(y - x)), y - x \rangle$$

for fixed $x, y \in \mathbb{R}^d$. Then:

$$f(y) - f(x) = \int_0^1 \phi'(t) dt = \int_0^1 \langle \nabla f(x + t(y - x)), y - x \rangle dt$$

This is useful for proving various properties:

- If we want to prove the tangent plane property from the monotonicity of the gradient:

$$\begin{aligned}
f(y) - f(x) &= \int_0^1 \langle \nabla f(x + t(y - x)), y - x \rangle dt \\
&= \int_0^1 \langle \nabla f(x + t(y - x)) - \nabla f(x) + \nabla f(x), y - x \rangle dt \\
&= \int_0^1 \langle \nabla f(x + t(y - x)) - \nabla f(x), y - x \rangle dt + \langle \nabla f(x), y - x \rangle \\
&\geq 0 + \langle \nabla f(x), y - x \rangle \quad (\text{by monotonicity of the gradient})
\end{aligned}$$

- If we want to prove that f is convex from $\nabla^2 f(x) \in S_d^+$ for all x :

$$\begin{aligned}
f(y) - f(x) &= \int_0^1 \langle \nabla f(x + t(y - x)), y - x \rangle dt \\
&= \int_0^1 \langle \nabla f(x + t(y - x)) - \nabla f(x) + \nabla f(x), y - x \rangle dt
\end{aligned}$$

Similarly as before, we can define $\varphi(t) = \langle \nabla f(x + t(y - x)), y - x \rangle$

$$\varphi'(t) = (y - x)^T \cdot \nabla^2 f(x + t(y - x)) \cdot (y - x) \geq 0 = \langle \nabla^2 f(x + t(y - x))(y - x), y - x \rangle$$

$$f(y) - f(x) = \int_0^1 \int_0^t \varphi'(s) ds dt + \langle \nabla f(x), y - x \rangle$$

Since $\nabla^2 f$ is positive semi-definite, we have $\varphi'(s) \geq 0$, which implies that $f(y) - f(x) \geq \langle \nabla f(x), y - x \rangle$.

2.1 General Convexity Exercises

Reminder: A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex if for all $x, y \in \mathbb{R}^d$ and $t \in [0, 1]$, we have

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y).$$

A function f is strongly convex with parameter $\alpha > 0$ if for all $x, y \in \mathbb{R}^d$ and $t \in [0, 1]$, we have

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y) - \frac{\alpha}{2}t(1 - t)\|x - y\|^2.$$

Exercise 2.1. Let $f \in \mathcal{C}^1(\mathbb{R}^d, \mathbb{R})$ be convex. Is it true that f has a unique critical point?

It is not necessarily true that a convex function has a unique critical point. As a counterexample, consider the convex function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = e^x.$$

This function is convex since its second derivative $f''(x) = e^x > 0$ for all $x \in \mathbb{R}$. However, it has no critical points because its first derivative $f'(x) = e^x$ is never zero. $\square$

Exercise 2.2. Let $f \in \mathcal{C}^1(\mathbb{R}^d, \mathbb{R})$ be convex. Show that if f has a critical point at $x^* \in \mathbb{R}^d$, then x^* is a global minimum of f .

If $\nabla f(x^*) = 0$ for some $x^* \in \mathbb{R}^d$, then for any $x \in \mathbb{R}^d$, the tangent plane property of convex functions gives us

$$f(x) \geq f(x^*) + \langle \nabla f(x^*), x - x^* \rangle = f(x^*) + 0 = f(x^*).$$

Thus, $f(x) \geq f(x^*)$ for all $x \in \mathbb{R}^d$, which means that x^* is a global minimum of f . $\square$

Exercise 2.3. Let $f \in \mathcal{C}^1(\mathbb{R}^d, \mathbb{R})$ be strictly convex. Show that f has at most one critical point.

Assume for the sake of contradiction that f has two distinct critical points x_1^* and x_2^* in $\mathbb{R}^d$. By the definition of critical points, we have $\nabla f(x_1^*) = 0$ and $\nabla f(x_2^*) = 0$. Since f is strictly convex, for any $\lambda \in (0, 1)$, we have

$$f(\lambda x_1^* + (1 - \lambda)x_2^*) < \lambda f(x_1^*) + (1 - \lambda)f(x_2^*).$$

However, since both x_1^* and x_2^* are critical points, we have

$$f(x_1^*) = f(x_2^*).$$

Thus,

$$f(\lambda x_1^* + (1 - \lambda)x_2^*) < f(x_1^*),$$

which contradicts the fact that both x_1^* and x_2^* are minima. Therefore, f can have at most one critical point. $\square$

2.2 Strong Convexity Exercises

Exercise 2.4. Show that f is strongly convex with parameter $\alpha > 0$, if and only if the function $g(x) = f(x) - \frac{\alpha}{2}\|x\|^2$ is convex.

The $\implies$ direction: Assume that f is strongly convex with parameter $\alpha > 0$. For any $x, y \in \mathbb{R}^d$ and $t \in [0, 1]$, we have

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y) - \frac{\alpha}{2}t(1-t)\|x - y\|^2.$$

Let's first deduce a magic formula to replace $\|tx + (1-t)y\|^2$

$$\begin{aligned}\|tx + (1-t)y\|^2 &= \langle tx + (1-t)y, tx + (1-t)y \rangle \\ &= t^2\|x\|^2 + (1-t)^2\|y\|^2 + 2t(1-t)\langle x, y \rangle.\end{aligned}$$

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2\langle x, y \rangle \implies \langle x, y \rangle = \frac{\|x\|^2 + \|y\|^2 - \|x - y\|^2}{2}$$

Plugging this into the $\|tx + (1-t)y\|^2$ expression, we get

$$\boxed{\|tx + (1-t)y\|^2 = t\|x\|^2 + (1-t)\|y\|^2 - t(1-t)\|x - y\|^2.}$$

Now for $g(tx + (1-t)y)$ we have:

$$\begin{aligned}g(tx + (1-t)y) &= f(tx + (1-t)y) - \frac{\alpha}{2}\|tx + (1-t)y\|^2 \\ &= \underbrace{f(tx + (1-t)y)}_{\text{Strongly convex}} - \frac{\alpha}{2}(t\|x\|^2 + (1-t)\|y\|^2 - t(1-t)\|x - y\|^2) \\ &\leq tf(x) + (1-t)f(y) - \frac{\alpha}{2}t(1-t)\|x - y\|^2 - \frac{\alpha}{2}(t\|x\|^2 + (1-t)\|y\|^2 - t(1-t)\|x - y\|^2) \\ &\leq tf(x) + (1-t)f(y) - \frac{\alpha}{2}(t\|x\|^2 + (1-t)\|y\|^2) \\ &\leq tg(x) + (1-t)g(y)\end{aligned}$$

The $\impliedby$ direction follows by assuming g is convex and reversing the steps above by using the magic formula for $\|tx + (1-t)y\|^2$.

Exercise 2.5 (Some examples of strongly convex functions). Determine which of the following functions are strongly convex on $\mathbb{R}^d$:

1. $f(x) = \|x\|^2$
2. $f(x) = e^{\|x\|}$
3. $f(x) = \frac{1}{2}x^T Ax + b^T x + c$, where A is a symmetric positive definite matrix.

Recall that a function f is strongly convex with parameter $\alpha > 0$ if for all $x, y \in \mathbb{R}^d$ and $t \in [0, 1]$, we have

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y) - \frac{\alpha}{2}t(1-t)\|x - y\|^2.$$

1. For $f(x) = \|x\|^2$, it is strongly convex with parameter $\alpha = 2$, because by the tangent plane property of strongly convex functions, we have

$$\|y\|^2 = \|x + y - x\|^2 = \|x\|^2 + 2\langle x, y - x \rangle + \|y - x\|^2$$

This shows that it is at most strongly convex with parameter 2.

2. The Hessian of $f(x) = e^{\|x\|}$ Calculate its gradient and Hessian we have

$$\nabla f(x) = e^{\|x\|} \frac{x}{\|x\|}$$

$$\nabla^2 f(x) = e^{\|x\|} \left(\frac{I}{\|x\|} + \frac{xx^\top}{\|x\|^2} - \frac{xx^\top}{\|x\|^3} \right)$$

$$\text{Eigenvalues of } \nabla^2 f(x) : \begin{cases} e^{\|x\|} & (\text{radial direction}) \\ \frac{e^{\|x\|}}{\|x\|} & (\text{any of the } n-1 \text{ orthogonal directions}) \end{cases}$$

We conclude that the function is strongly convex on $(x \neq 0)$ with parameter

$$\alpha = \begin{cases} e^{\|x\|} & \text{if } \|x\| \geq 1 \\ \frac{e^{\|x\|}}{\|x\|} & \text{if } 0 < \|x\| < 1 \end{cases}$$

3. For $f(x) = \frac{1}{2}x^T A x + b^T x + c$, where A is a symmetric positive definite matrix, the Hessian of f is given by $\nabla^2 f(x) = A$. Since A is positive definite, there exists a constant $\alpha > 0$ such that for all $x \in \mathbb{R}^d$,

$$x^T A x \geq \alpha \|x\|^2.$$

Therefore, f is strongly convex with parameter α . □

Exercise 2.6. Let $f \in \mathcal{C}^2(\mathbb{R}^d, \mathbb{R})$ be α -strongly convex. Does f admit a unique minimizer?

By the strong convexity of f , we know that under the assumption of the existence of a minimizer, it must be unique. Now we show the coercivity of f to ensure the existence of a minimizer.

One way to do this is to show that $f(x) \rightarrow +\infty$ as $\|x\| \rightarrow +\infty$. Since f is α -strongly convex, by the tangent plane property, we have for any $x, x_0 \in \mathbb{R}^d$:

$$f(x) \geq f(x_0) + \langle \nabla f(x_0), x - x_0 \rangle + \frac{\alpha}{2} \|x - x_0\|^2.$$

For the sake of the clarity, let $x_0 = 0$, we have

$$f(x) \geq f(0) + \langle \nabla f(0), x \rangle + \frac{\alpha}{2} \|x\|^2.$$

This is a quadratic function in $\|x\|$ with a positive leading coefficient $\frac{\alpha}{2}$, which implies that $f(x) \rightarrow +\infty$ as $\|x\| \rightarrow +\infty$. Therefore, f is coercive. □

A more rigorous way is to use the sublevel set definition: a function f is coercive if the sublevel sets $\{x \in \mathbb{R}^d : f(x) \leq c\}$ are bounded for all $c \in \mathbb{R}$.

$$f(x) \geq f(0) + \langle \nabla f(0), x \rangle + \frac{\alpha}{2} \|x\|^2$$

Recall that Young's inequality states that for any $a, b \geq 0$ and $\epsilon > 0$, we have

$$ab \leq \frac{a^2}{2\epsilon} + \frac{\epsilon b^2}{2}.$$

Now we control the linear term by the quadratic term using Cauchy-Schwarz and Young's inequality (with $\epsilon = \frac{\alpha}{2}$):

$$\langle \nabla f(0), x \rangle \geq -\|\nabla f(0)\| \|x\| \geq -\frac{1}{\alpha} \|\nabla f(0)\|^2 - \frac{\alpha}{4} \|x\|^2$$

Plugging this back we have

$$f(x) \geq f(0) - \frac{1}{\alpha} \|\nabla f(0)\|^2 + \frac{\alpha}{4} \|x\|^2$$

Thus the sublevel sets are bounded. Therefore, f is coercive. □

3 Lipschitz Continuity of the Gradient

1. **Definition:** A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ has a Lipschitz continuous gradient with constant $\mu > 0$ if for all $x, y \in \mathbb{R}^d$, we have:

$$\|\nabla f(y) - \nabla f(x)\| \leq \mu \|y - x\|$$

2. **Several Implications for Lipschitz Continuity:** Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a differentiable function with a Lipschitz continuous gradient with constant $\mu > 0$. Then for all $x, y \in \mathbb{R}^d$, we have:

- Upper bound on the Hessian eigenvalues:

$$\lambda_1(\nabla^2 f(x)) \leq \mu$$

- Quadratic upper bound using gradient at x :

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2} \|y - x\|^2$$

4 Caratheodory theorem

1. **Convex hull:** The convex hull of a set $\Omega \subset \mathbb{R}^d$ is defined as the smallest convex set containing Ω .

$$\text{conv}(\Omega) := \bigcap \{C \subset \mathbb{R}^d \mid C \text{ is convex and } \Omega \subset C\}$$

It can be expressed as:

$$\text{conv}(\Omega) = \left\{ \sum_{i=1}^N \lambda_i x_i \mid N \in \mathbb{N}, x_i \in \Omega, \lambda_i \geq 0, \sum_{i=1}^N \lambda_i = 1 \right\}$$

2. **Caratheodory theorem:** Let $\Omega \subset \mathbb{R}^d$. Then any point in the convex hull of Ω can be expressed as a convex combination of at most $d + 1$ points from Ω . In other words, for any $x \in \text{conv}(\Omega)$, there exist points $x_1, x_2, \dots, x_{d+1} \in \Omega$ and coefficients $\lambda_1, \lambda_2, \dots, \lambda_{d+1} \geq 0$ such that:

$$x = \sum_{i=1}^{d+1} \lambda_i x_i \quad \text{and} \quad \sum_{i=1}^{d+1} \lambda_i = 1$$

5 Extreme Points and Krein-Milman Theorem

1. **Extreme Points:** Let C be a convex set in a vector space. A point $x \in C$ is called an extreme point of C if **it cannot be expressed as a convex combination of two distinct points in C** . Formally, x is an extreme point if whenever $x = \lambda y + (1 - \lambda)z$ for some $y, z \in C$ and $\lambda \in (0, 1)$, it follows that $y = z = x$.

- 2. Projection on closed convex sets:** Let $C \subset \mathbb{R}^d$ be a non-empty closed convex set. For any point $x \in \mathbb{R}^d$, there exists a **unique** point $P_C(x) \in C$ such that:

$$\|x - P_C(x)\| = \min_{y \in C} \|x - y\|$$

The point $P_C(x)$ is called the projection of x onto the set C . The projection operator $P_C : \mathbb{R}^d \rightarrow C$ is **1-Lipschitz** (non-expansive), meaning that for all $x, y \in \mathbb{R}^d$:

$$\|P_C(x) - P_C(y)\| \leq \|x - y\|$$

- 3. Krein-Milman Theorem:** Let C be a non-empty compact convex subset of a locally convex topological vector space. Then C is the closed convex hull of its extreme points. In other words, if we denote the set of extreme points of C by $\text{ext}(C)$, then:

$$C = \overline{\text{conv}}(\text{ext}(C))$$

where $\overline{\text{conv}}(\text{ext}(C))$ denotes the closure of the convex hull of the extreme points of C .

In other words, every compact convex set can be "reconstructed" from its extreme points by taking all possible convex combinations of those points and then taking the closure of that set.

6 Polyak-Lojasiewicz Inequality

- 1.** Let's remind first the definition and some properties of strongly convex functions. A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is called α -strongly convex if for all $x, y \in \mathbb{R}^d$:

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\alpha}{2} \|y - x\|^2 \quad (\text{Function is above the Tangent Cone})$$

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq \alpha \|y - x\|^2 \quad (\text{Curvature Condition})$$

- 2. Polyak-Lojasiewicz Inequality:** A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is said to satisfy the Polyak-Lojasiewicz (PL) inequality with constant $\mu > 0$ if for all $x \in \mathbb{R}^d$:

$$\frac{1}{2} \|\nabla f(x)\|^2 \geq \mu(f(x) - f^*)$$

where $f^* = \inf_{x \in \mathbb{R}^d} f(x)$ is the global minimum value of f .

- 3.** We can show that α -strongly convex functions satisfy the PL inequality with constant $\mu = \alpha$. Let f be an α -strongly convex function and let x^* be its unique minimizer. Then for any $x \in \mathbb{R}^d$:

$$\begin{aligned} f(x) &\geq f(x^*) + \langle \nabla f(x^*), x - x^* \rangle + \frac{\alpha}{2} \|x - x^*\|^2 \\ &\geq f(x^*) + 0 + \frac{\alpha}{2} \|x - x^*\|^2 \quad (\text{since } \nabla f(x^*) = 0) \\ \Rightarrow f(x) - f(x^*) &\geq \frac{\alpha}{2} \|x - x^*\|^2 \end{aligned}$$

On the other hand, making a minor modification to the first property of strongly convex functions, we have:

$$\begin{aligned}
f(x) - f(y) &\leq \langle \nabla f(x), x - y \rangle - \frac{\alpha}{2} \|x - y\|^2 \\
&\leq \|\nabla f(x)\| \|x - y\| - \frac{\alpha}{2} \|x - y\|^2 \quad (\text{by Cauchy-Schwarz inequality}) \\
&\leq \frac{\|\nabla f(x)\|^2}{2\alpha} \quad (\text{by maximizing the right-hand side over } \|x - y\|)
\end{aligned}$$

Combining the two inequalities, we get:

$$f(x) - f(x^*) \leq \frac{\|\nabla f(x)\|^2}{2\alpha}$$

Rearranging this gives the PL inequality:

$$\frac{1}{2} \|\nabla f(x)\|^2 \geq \alpha(f(x) - f(x^*))$$